MILITARY SPECIFICATION

COMPUTER, HELICOPTER PERFORMANCE

This specification is approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers the requirement for a helicopter performance computer of the circular disc type for piloted helicopter aircraft. The computer described herein is of three disc construction, hand-held, pocket size, and applies to the aircraft designated thereon.

2. APPLICABLE DOCUMENTS

2.1 The following publications, of the issue in effect on date of invitation for bid form a part of this specification to the extent specified herein.

2.1.1 Specifications -

Federal

PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive, Adhesive, Waterproof for Packaging and Sealing
PPP-B-676	Boxes, Paperboard, Setup
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)

Military

MIL-B-4229	Boxes, Paperboard, Metal-Stayed
MIL-D-70327	Drawings, Engineering and Associated Lists
MIL-H-5474A	Handbooks and Breakdowns; general preparation of
MIL-STD-105	Sampling Procedures and Tables for Inspection Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, (Code CDS), Philadelphia 20, Pennsylvania).

3. REQUIREMENTS

3.1 Standard Sample - A computer sample will be furnished by the Bureau of Naval Weapons Representative (see paragraph 6.1) which can be used as a design guide.

FSC 6605

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3.2 Design - The computer shall be designed to provide hover and cruise information for piloted helicopter aircraft. The design, format, and fabrication details shall be in accordance with the requirements of this document and generally in accordance with Figures 1 and 2.

3.2.1 Basis for Computer Data - Data used for the computer shall be the most reliable available at the time of computer issue. The performance data used in preparing the computer design shall be that data derived from flight tests whenever available, otherwise, conservative estimates shall be used until verified by flight tests. To justify the basis of data as FLIGHT TEST, all performance information incident to the determination of the data entered on the computer shall be based on the results of flight tests.

3.2.1.1 Supporting Data - Data supporting the information contained on the computer shall be submitted to the Chief, Bureau of Naval Weapons concurrently with the preproduction computer samples. These data may be in the form of copies of the aircraft performance curves used for the computer design. These data will be used to evaluate the information contained on the preproduction computer samples (see paragraph 3.5).

3.2.2 Aircraft Weight - The computer shall be designed to provide information for the complete range of aircraft weights from maximum gross to empty, and the designed loading configurations.

3.3 Construction - The computer shall consist of three (3) concentric discs and a circular segment permanently assembled on a common pivot. The base disc shall have one (1) rotatable disc and one (1) segment mounted on the front face (Figure 1) and one (1) rotatable disc mounted on the back face (Figure 2).

3.3.1 Discs - The discs of the computer shall be fabricated of plastic materials. The basic disc shall be opaque; the other discs and segment transparent except for screened and printed areas.

3.3.1.1 Thickness of Discs - The base disc shall be of 0.040 ± 0.004 inch thickness. The other discs and segment shall be of 0.030 ± 0.003 inch thickness.

3.3.1.2 Tabs on Discs - A circular finger tab shall be incorporated on the rotatable disc of the back face (Figure 2). A finger shall protrude approximately 1/8 inch from the segment (Figure 1).

3.3.1.3 Disc Laminations - All printing and screening applied to plastic shall be laminated with a transparent plastic sheet of 0.002 inch minimum thickness and be sealed against the entrance of moisture between the laminations. To insure proper sealing, no screening or printing shall be closer than one eighth inch from the disc edges.

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3.3.1.4 Diameter of Discs -

a. The diameter of the base disc shall be 6± 1/16 inches.

b. The diameter of the back face top disc shall be the same as the base disc including a transparent rim area.

c. The diameter of the front face middle disc shall be approximately 5 1/4 inches including a transparent rim area. The edge of the disc shall not obscure the scale or scale figures.

d. The radius of the segment shall be approximately 2 9/16 inches to meet the mating scale of the base disc. The segment edge shall not obscure the mating scale.

3.3.1.5 Disc Windows - All clear windows in the discs shall be approximately dimensioned so as not to obscure the markings, letters and numbers intended to be seen through the windows.

3.3.1.6 Markings - All markings, which include letters and numbers, shall be black on white, white on black, or black on colors and shall be sharply defined and legible in both white light or daylight and under the red lighting of the aircraft cockpit. Offset printing is acceptable. Scale markings, lettering, and tabulations shall conform in general to the size, arrangement, and content to that indicated in Figures 1 and 2.

3.3.2 Assembly - A flat plastic washer of 1/16 inch nominal thickness and 1/2 to 3/4 inch diamater shall be inserted between each pair of discs in the assembly concentric with the pivot axis. A metal eyelet and spring washer shall be used in assembling the discs concentrically. The eyelet shall be of suitable length and 3/16 inch nominal outside diameter. The spring washer shall be of 1/2 ± 1/16 inch diameter and shall be assembled on the back face of the computer.

3.3.2.1 Movement - The discs shall move easily and smoothly and yet have sufficient friction to retain their position in normal use.

3.4 Materials -

3.4.1 Plastic - The plastic material used shall be vinyl chloride polymer or copolymer resin; shall be homogeneous, flat, and of a type having maximum resistance to warpage.

3.4.2 Metal Fittings - The eyelet and spring washer used shall be of the corrosion resistant type, unless suitably protected to resist corrosion during normal service life. They shall be of adequate strength to resist the normal forces resulting from handling and manipulating the computer.

3.5 Preproduction Sample - For each issue of a computer of a new part number two (2) preproduction sample computers from final printing plates intended for production shall be furnished by the contractor for approval prior to commencement of computer production. The criteria for assignment of a new part number are established by paragraph 3.5.4 of MIL-D-70327.

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3.6 Accuracy - The accuracy with which values may be obtained with the computer is contingent upon the accuracy of data used in design of the computer, the accuracy with which the master layouts are made and the discs printed, and the accuracy with which the discs are assembled into a finished computer. The computer shall be checked for accuracy in the manner specified in Section 4.

3.7 Finish - All faces shall have a smooth surface. The disc faces may all have a dull finish, however, a high degree of transparency shall be provided in clear areas through which markings appear.

3.8 Identification of Product -

3.8.1 Product Labelling - The following information shall be permanently and legibly marked on each computer:

Aircraft Model(s)
Secruity Classification
COMPUTER, HELICOPTER PERFORMANCE
FSN (Federal Stock Number)
Contractor's Part No.
Contract No.
Contractor's Name or Trade Mark
U. S. Property
Issue Date
Type Data (Estimated or Flight Test)

3.9 Operating Instructions - Operating instructions shall be included on the face of the computer. If feasible a solution of a typical example should also be included. Otherwise, an instruction pamphlet shall be prepared, reproduced, and one packaged with each computer including the two (2) pre-production computers (paragraph 3.5). The instruction pamphlet shall be prepared in accordance with Specification MIL-H-5474A insofar as format, page size, punched holes and type size are concerned.

3.10 Workmanship - The computer, including all parts, shall be free of burrs, sharp edges or any other evidence of poor workmanship.

3.11 Disc Test Markings - Inconspicuous circular arcs, shown in Figures 1 and 2, shall be made near the outer diameter of each disc for the purpose of determining disc reproduction and assembly accuracies (see paragraphs 4.3.2 and 4.3.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. The supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Sampling - The sampling plans and procedures to be used in the determination of the acceptability of products shall be in accordance with MIL-STD-105, inspection level II, Acceptable Quality Level (AQL) 1.0 percent.

4.1.2 Preproduction Sample - Two (2) computers shall be manufactured and submitted to the inspector who shall examine the samples to verify strict conformance with drawings and specifications prior to preproduction tests at the Bureau of Naval Weapons. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Submittal of Preproduction Samples - Sample computers shall be submitted for test and evaluation to the Chief, Bureau of Naval Weapons (RAPP) plainly marked on the envelope as in paragraph 5.3.1 plus the following:

Sample for Preproduction Test
submitted by (supplier) (date) for
preproduction test in accordance with the
requirements of Specification MIL-C-22869(Wep)

4.1.3 Lot Size - As applied to Government inspection of units or product the term "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units or product in "inspection lots" will be determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Classification of Inspection -

(a) Acceptance inspection
(b) Preproduction inspection

4.3 Acceptance Inspection - These inspections are to be accomplished on each Computer being inspected under the sampling plan of 4.1.1.

4.3.1 Examination of Product - Each Computer to be inspected under the sampling plan of 4.1.1 shall be inspected to determine conformance with the requirements of 3.3, 3.4, 3.7, 3.8, 3.9, 3.10 and Section 5. Non-conforming Computers shall be governed by paragraph 4.5.

4.3.2 Disc Inspection - The accuracy with which the master layouts are reproduced shall be determined by measuring the diameter of the circular segment markings (paragraph 3.11) to determine that the diameter does not vary more than 1/64 inch from the intended diameter. These measurements may be made prior to assembly of the Computer.

4.3.3 Assembly Accuracy - Upon completion of assembly, the accuracy with which the discs and segment are assembled shall be determined by determining visually with reference to the circular segment marks (paragraph 3.11) that eccentricity does not exceed 1/64 inch.

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4.4 Preproduction Inspection - Preproduction inspection shall consist of all the inspections contained in this specification.

4.5 Disposition of nonconforming products - Disposition of nonconforming Computers shall be in accordance with MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A

5.1.1.1 Unit Package - Each Computer shall be enclosed, along with a copy of the instruction pamphlet if prepared, in a kraft envelope having a bursting strength (Mullen) of 56 pounds per square inch, end opening, with gummed flap. Size of envelope shall be 7" x 7" or sufficient to close flap with Computer, or with Computer and pamphlet as applicable, in the envelope.

5.1.1.2 Intermediate Packaging - Computers shall be packaged in an intermediate container for all shipments except for domestic shipments of fewer than twenty (20) units to an addressee. The Computers shall be packaged in a close fitting manner, with not more than twenty-five (25) units within a setup box in accordance with Specification PPP-B-676 or a metal-stayed box in accordance with Specification MIL-B-4229 using gummed tape conforming to Specification UU-T-111.

5.1.2 Level C

5.1.2.1 Packaging - Unit packaging shall be the same as for Level A, (see paragraph 5.1.1.1). Other packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing -

5.2.1 Level A - Intermediate packages, prepared as specified in 5.1.1.2, shall be packed in an overseas type fiberboard box in accordance with Specification PPP-B-636, sealed with tape in accordance with Specification PPP-T-60.

5.2.2 Level B - Intermediate packages, prepared as specified in 5.1.1.2, shall be packed in a domestic type fiberboard box in accordance with Specification PPP-B-636, sealed with tape in accordance with Specification PPP-T-60.

5.2.3 Level C - Unless otherwise specified, interior packages shall be packed in substantial commercial shipping containers constructed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of Uniform Freight Classification in effect at the time of shipment, except that the use of corrugated or solid fiberboard, having a minimum Mullen test of less than 275 pounds is prohibited.

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5.3 Marking and Labeling -

5.3.1 Special Markings - None.

5.3.2 Normal Markings - In addition to markings required by contract or order, unit package, intermediate package, and outer container shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Standard Sample - A sample computer can be obtained from the Bureau of Naval Weapons Representative. Unless otherwise specified, a computer for the airplane most similar should be chosen for the sample.

6.2 Computer Improvement - Suggestions for improvement of the computer should be made to the Chief, Bureau of Naval Weapons.

6.3 Ordering Data - Requisitions, contracts, and orders should specify the level of packaging and number of unit and intermediate containers per pack.

6.4 Flight Testing - Flight testing is not authorized by this specification.

6.5 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

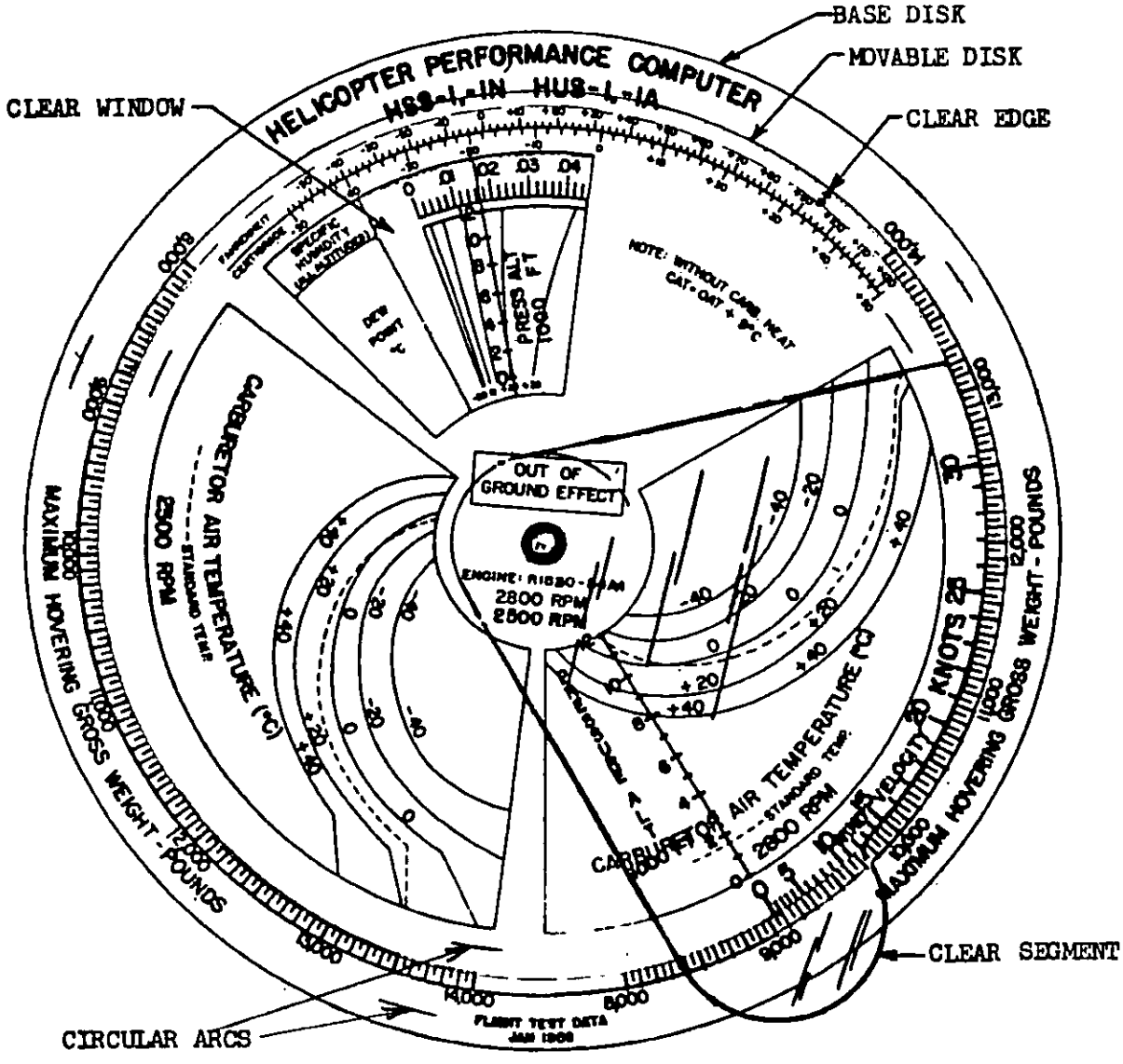
For Level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

NOTICE: When Government drawings, specification, or other data are used for any purpose other than in connection with definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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HELICOPTER PERFORMANCE COMPUTER
FRONT FACE

FIGURE 1

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